

CS 407(CS 503) Theory of Computation - Spring 2023

	<i>NAME/SCHEDULE</i>	<i>EMAIL/TEL</i>	<i>OFFICE HOUR</i>
<i>INSTRUCTOR</i>	<i>Kemal İNAN</i>	<i>inan</i>	<i>by appointment</i>
<i>ASSISTANT(S)</i>			
<i>LECTURES</i>	<i>Tu 10:40 – 12:30 F 13:40 – 14:30</i>		
<i>RECITATION</i>	<i>T 17:40 – 18:30</i>	<i>FENS L062</i>	

***Main Text : H. R. Lewis & C. H. Papadimitriou , Elements of Theory of Computation,
2nd ed. Prentice Hall 1998 (out-of-print; e-copy available)***

Auxiliary Texts :

(1)M. Sipser, Introduction to the Theory of Computation, 3rd ed., Cengage Learning 2003

(2) M. A. Garey& D. S. Johnson , Computers and Intractability, Bell Telephone Labs 1979

Grading: 25% midterm , 35% quizzes, 40% Final

***A total of 10 quizzes each of 20 minutes duration will be given during chosen class
lectures ; minimum 7 quiz entries are required for a passing grade irrespective of health
or similar legitimate-looking excuses.***

25 =

Tentative Spring 2023 Schedule

(Tuesday 2 hour courses are encoded in green color)

Feb/March	17 1,2	20 2	24 2,3 HW1	27 3	3 3,4	6 4	10 4 HW2	13 4,5
March/April	17 5,6	24 6,7	27 7,8 HW3	31 8	3 8	7 8,9 HW4	10 9	14 9,10
April/May	17 Midterm	21 11	24 11 HW5	28 12,13	5 13	8 14	12 14,15 HW6	15 15
May	19 15,16 HW7	22 16						

TENTATIVE COURSE OUTLINE

1- Turing Machines : Definition, Representations and Computational Concepts

(Main Text 4.1- 4.2) S1

2 – Extended Turing Machines : Multitape, RAM Machine and Equivalences

(MT 4.3 – 4.4) S2

3 – Nondeterministic TM (MT 4.5) S2

4 – Grammars and Computation (MT 4.6) S3

5 – Numerical functions : primitive recursion and μ - recursion (MT 4.7) S3

6 – Computational equivalence of grammars, TMs and recursive functions (MT 4.7) S3

7– Up-down counter and Universal Turing Machines (MT Prob. 5.4.5, 5.2) S4

8– Decidability: Church Turing thesis and the Halting Problem (MT 5.3) S5

9 – Reducability and Unsolvable Problems (MT 5.4 – 5.7) S5

10 – Recursion (Sipser, Chapter 6 , pp. 197-203) S5

11 – Computational Complexity: Concepts and Definitions S6

(MT 6.1 – 6.3 &Garey and Johnson 3.1)

12 – The Class P and NP (MT 6.4) S6

13 – Polynomial Reductions and NP-Completeness (MT 7.1) S6

14 – Cook’s Theorem (MT 7.2 &Garey 2.6) S6

15 – Some NP- Complete Problems (MT 7. &Garey 3.1) S6

16 – Space Complexity (Sipser, Chapter 8 selections) S6